

Multi-Person Localization from UWB Radar CIR

A TinyML occupancy-grid system for the ESP32-S3

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1 Problem and constraints

The system estimates the 2D positions of up to four people in a $4.8 \text{ m} \times 7.2 \text{ m}$ room from raw UWB radar channel impulse responses: six radars, three antennas each, 120 range bins, complex $\frac{I}{Q}$, sampled at 25 Hz. The trained model must run on an ESP32-S3 through TensorFlow Lite for Microcontrollers, under a 800 KB flash budget, a 300 KB activation arena, and a blocklist that rules out every recurrent operator. Scoring is global micro-averaged F_1 with Hungarian matching at a 1.0 m radius.

Final evaluation uses a separate acquisition session from the one used during development, so generalization across sessions — not just across held-out frames of the same recordings — is the relevant test of this system. All validation figures in this report use a leave-window-out cross-validation estimate rather than a same-session holdout, reported with its uncertainty.

2 Signal path

2.1 Decluttering: an EMA as a 0.04 Hz high-pass

A static room dominates the raw CIR. We subtract an exponential moving average of the signal from itself, $y_t = x_t - \text{EMA}_\alpha(x)_t$ with $\alpha = 0.99$, which is a first-order high-pass with a corner near 0.04 Hz. Everything stationary cancels; everything that moves survives.

Two details are load-bearing and neither is obvious:

- **The filter state is seeded from frame 0**, not from zero. Initialized at zero, the filter spends its first seconds reporting the entire static room as one enormous movement. `scipy.signal.lfilter_zi` gives the steady state for a constant input; scaling it by the first frame starts the filter as if the scene had been frozen there forever.
- **The streaming and batch implementations use the same code path.** Deployment applies `lfilter` to one frame at a time, carrying its filter state between calls, which reproduces the batch computation bit-for-bit. A more direct-looking recursion, $bg = \alpha \cdot bg + (1 - \alpha) \cdot x$, is not equivalent: its arithmetic order differs from `lfilter`'s internal implementation, and the two diverge by $\sim 4 \times 10^{-2}$ over a recording.

One direct consequence of this, which we confirmed rather than assumed: because the EMA seeds from frame 0, the decluttered frame 0 is identically zero in every recording. That is a cold-start artifact, and it reappears in Section 6.

2.2 Range-bin crop and the magnitude channel

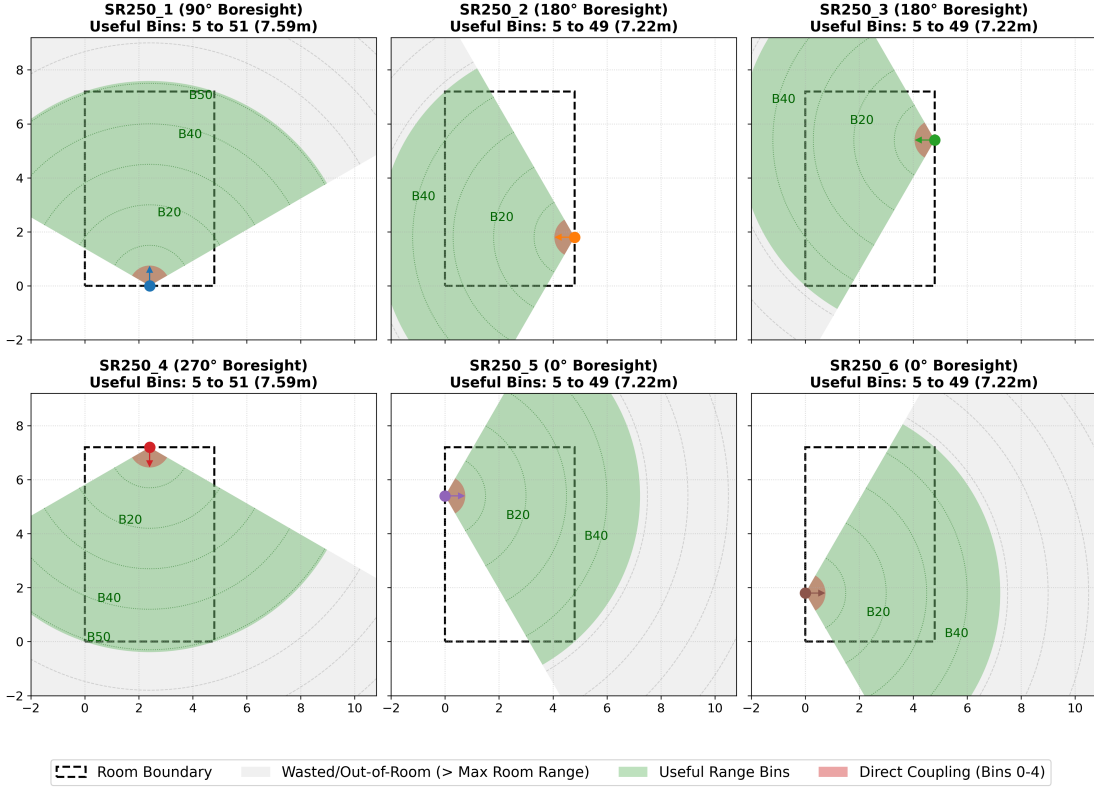


Figure 1: Coverage of the six radars over the $4.8 \text{ m} \times 7.2 \text{ m}$ room, and the mapping from range bins to physical distance. The crop to bins 5–52 removes only the sensor’s own coupling and the bins beyond the room.

Bins 0–4 carry transmit-to-receive coupling inside the sensor itself; bin 52 corresponds to roughly 7.8 m of range, just short of the room’s 8.65 m diagonal, so the crop keeps essentially the full room while excluding bins that can only be multipath. Cropping to [5:52] removes 61% of the input.

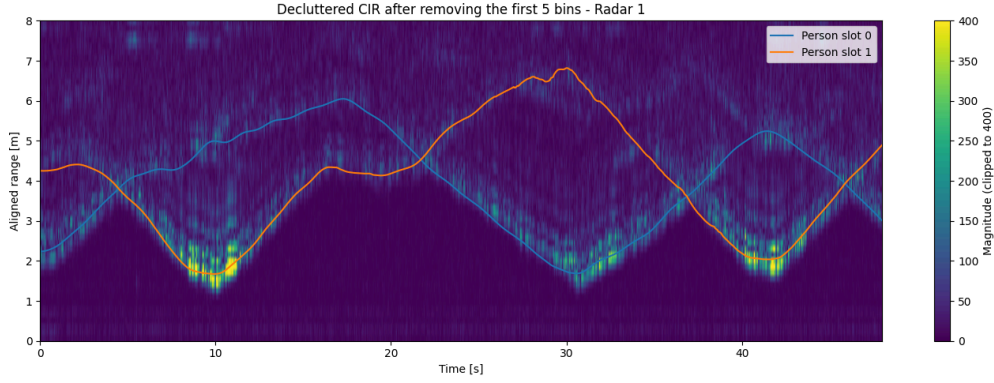


Figure 2: Decluttered magnitude for a single radar and antenna over one recording, with two tracked person trajectories overlaid — this is close to the representation the network actually receives, after the magnitude and normalization steps below.

The model is then fed $|I + jQ|$, normalized by a per-channel 99th percentile computed **on each fold’s training windows only**. One subtlety survives into the code as a comment: the magnitude is computed as `sqrt(re**2 + im**2)` and never as `np.abs` on a complex array, because `np.abs` uses a hypot-style algorithm whose result differs in the last bits. The cached training data and the deployed model were both built on the `sqrt` form, and mixing them silently breaks byte-level parity between training and inference.

Putting the chain together: the raw complex CIR is decluttered (§2.1), cropped to the 47 informative range bins, reduced to magnitude, and clipped and rescaled to $[0, 1]$ by the per-fold 99th percentile. The deployed model uses this single magnitude channel per radar and antenna; Section 7 describes seven additional channel sets that were tried against it and rejected.

3 Model architecture

Why the first layer is a Dense, not a convolution. The input axis of length 6 indexes **radars**, which are physically distributed around the room. A convolution over that axis would assert that radar 1 and radar 2 are neighbours in a way radar 1 and radar 6 are not — a translation-invariance claim about an axis that has no translation structure at all. The geometric mapping from six range-profile sets to one room is arbitrary and global, so it is learned by a fully-connected projection.

The network flattens the $(6, 3, 47, 1)$ input and projects it through a 256-unit Dense layer, then a second, 432-unit Dense layer whose output reshapes to an $18 \times 12 \times 2$ spatial feature map. Three depthwise-separable convolution blocks refine that map before a linear 1×1 head produces the final heatmap. Targets are Gaussian blobs ($\sigma = 0.3$ m) at each person’s position, decoded back by local-maximum extraction with sub-cell centre-of-mass refinement and non-maximum suppression.

Capacity is spent almost entirely in the early fusion: 330,851 parameters, of which 216,576 belong to the first Dense layer alone. Adding a second input channel costs another 216,576 and a third exceeds the flash budget outright — which is why Section 7’s channel experiments trade against the radar count.

3.1 Operator count: BatchNorm folding

The original blocks were `SeparableConv2D(activation="relu")` followed by `BatchNormalization`. TFLite folds a BN into the preceding operator’s weights only when the path between them is purely linear, so a ReLU sitting in between blocks the fold in both directions and three BNs shipped as runtime MUL+ADD pairs. Reordering to `Linear → BatchNorm → ReLU`, with the nonlinearity as its own layer, folds all three.

Separately, exporting with a **fixed batch-1 input signature** lets the converter constant-fold the reshape. With a dynamic batch dimension Keras must build the reshape target as `[tf.shape(x)[0], 18, 12, C]`, which ships as `SHAPE → STRIDED_SLICE → PACK → RESHAPE` where one static `RESHAPE` would do.

Together these took the deployed graph from **20 operators to 11**, with no change to the model itself.

4 Quantization

The calibration set matters more than anything else in the conversion, and it only became true after the BN fold. Before folding, three live MUL+ADD pairs rescaled activations at every inference, and that rescaling silently absorbed a badly-chosen calibration range. Once BN folds into the convolution weights, that safety net is gone: the range picked at conversion time is the only thing setting int8 resolution.

The original pipeline calibrated on **raw**, non-decluttered frames — which had been measured, pre-fold, to beat decluttered calibration. Post-fold, the same choice was catastrophic: raw frames are far larger in magnitude than decluttered ones, so the calibration range was set far too wide, and the quantized model produced **5,569 false positives** on a routine post-conversion check. Switching to decluttered calibration frames — the distribution the model actually trains and deploys on — resolved it.

This is the clearest case in the project of an optimization being validated by the wrong metric: the operator-count work succeeded on its own terms and would have shipped a badly broken model if the check had stopped at “op count went down”.

5 Validation methodology

5.1 Grouped k-fold, never frame-level

At 25 Hz, adjacent frames are near-duplicates. A frame-level split therefore leaks almost perfectly and inflates F_1 without measuring anything. All validation is by **whole acquisition window**: six folds, stratified by the number of people present, with each window scored by a model that never trained on it, and with the per-fold p_{99} and energy gate computed on that fold’s training windows only.

Detections are pooled across every fold and scored once, micro-averaged over the full set of windows — never averaged per window.

5.2 Reporting the interval, not the point

A leave-one-window-out jackknife over the 24 windows puts the standard error of pooled F_1 at ± 0.018 . Several changes considered during the project moved F_1 by less than a fifth of that.

For A/B comparisons the relevant quantity is smaller: two models scored on the same windows form a **paired** comparison, per-window difficulty cancels, and the SE of the difference is roughly half as large — ± 0.0089 for pooled F_1 , ± 0.0183 for seated recall, and ± 0.0047 for walking recall. We adopt a change only when its effect exceeds one paired SE, compared at **matched operating points**: an intervention scored at its own tuned threshold against a baseline at an untuned one is not a measurement.

5.3 Why there is no lockbox

Holding out three windows would carry $SE \approx 0.018\sqrt{24/3} \approx 0.05$ — far too coarse to adjudicate the 0.01-scale decisions actually being made, and it would guard the wrong risk. All 24 windows are one acquisition session; the real test is a different one, so a same-session holdout resembles training far more than it resembles the test. The 12.5% of data would buy false confidence. We report the interval and hold the adoption bar instead.

6 Conditional inference

A per-frame model spends the same computation on an empty room as on four walking people, but after decluttering, an empty room is near-silent. We use this: the mean of the normalized magnitude over all 846 input cells is a single scalar that distinguishes an occupied frame from an empty one. When it falls below a calibrated gate, the model is not invoked at all, and the all-zero heatmap it would have produced is substituted directly. The gate itself costs 846 additions, against roughly 330,000 multiply-accumulates for a full inference — a skipped frame costs about 0.3% of the compute a normal one would.

Calibrating this gate produced two findings.

The statistic that separates classes better makes a worse gate. Reducing each frame to its **maximum** rather than its mean more than doubles the pooled separability ($d' = 5.53$ against 2.76) — but at a matched safety margin, it skips only 77.0% of empty-room frames, against 95.1% for the mean. The maximum’s higher d' comes from walking frames saturating at 1.0, which reflects how bright a walker is, not where the empty-versus-occupied boundary actually sits. A statistic measured far from the decision boundary does not predict behaviour at it.

“Never skip an occupied frame” is not a calibratable rule on this dataset. Window 021 holds a single seated subject with nobody else moving, and its occupied frames are **dimmer than an empty room** — median frame energy 0.01693 against 0.01627 and 0.01632 for the two genuinely empty recordings, a ratio of 1.04. A gate that protects it must sit below every empty room as well, and then skips 0.06% of frames: the feature buys nothing.

The gate is therefore calibrated against the windows where a person is separable from an empty room **at all** — operationally, those whose 1st-percentile occupied energy exceeds the pooled empty-room median — and set to a 0.9 margin below the dimmest such frame. Window 021 is excluded by that measurement rather than by name, so the rule survives a dataset in which a different window is the pathological one. The result skips **95.1% of empty-room frames and 0.0000% of occupied frames across all 21 separable windows**.

Frame 0 is always gated, in every recording, and correctly so: the decluttering EMA seeds from it, so the decluttered frame 0 is identically zero and the model has nothing to see there either.

This gate is not only a compute saving. Because it suppresses the model precisely where phantom detections are most likely — a genuinely empty room — it functions as a quality intervention as well.

7 The seated-subject investigation

The single largest error mode in the project: eight of the 24 windows contain a seated subject, and they hold roughly three quarters of all pooled false negatives. Window 021, seated-only, scored $F_1 \approx 0.006$.

7.1 The signal is present; the per-frame representation is what fails

Within the decluttered complex CIR, energy in the 0.15–0.6 Hz respiration band is 52–168× the energy at other frequencies in every window containing at least one person — against only 1.8–1.9× in the two genuinely empty recordings. A respiration signal is therefore measurably present whenever anyone is in the room, seated or not. But a **single magnitude frame** separates seated-from-empty at only $d' = 1.44$, against $d' = 2.45$ for the envelope of the same band measured over time. What is missing is temporal context, not phase.

7.2 Eight attempts, all negative

Channel set	Idea	Result
mag+delta32	complex difference at a 1.28 s lag	pooled F_1 +0.6 SE (noise)
mag+aoa	phase difference between radars	pooled F_1 -2.4 SE (regression)
iq	raw $\frac{I}{Q}$ instead of magnitude	pooled F_1 -0.1 SE (noise)
mag+mag27+55+82	four instants across one breath	F_1 -9.1 SE vs. 6 radars; -0.5 SE once radar count was controlled for
oversampling ×3	seated windows repeated in training	pooled F_1 -1.7 SE; seated recall +0.8 SE, below the adoption bar
mag+breath	causal 0.15–0.6 Hz respiration bandpass	seated recall +1.2 to +2.1 SE; walking recall -5.3 to -10.4 SE
mag+perio (raw)	periodicity at the respiration lag	ruled out before training by its own pre-run check

Channel set	Idea	Result
mag+perio (coherence)	amplitude-invariant form of the above	pooled F_1 -8 to -11 SE, the largest regression

The two periodicity-based channels are described together because they failed for **opposite** reasons, and together close off the approach rather than simply adding one more negative result. In its raw form the statistic is mathematically equal to $|A|^2 \times \text{coherence}$ — reflector strength squared, times how periodic the signal is — so it behaves as an energy measurement more than a periodicity one, and a walker’s much larger reflection amplitude beats a breather’s genuine periodicity by roughly 400×. Dividing out that amplitude fixes this, but removes the one thing that let the statistic recognize an empty cell: even pure noise, averaged the same way over a finite window, does not settle to zero — it settles near $1/\sqrt{N_{\text{eff}}}$, a small but real residual. Every empty cell (845 of the 846 in any given frame) then reads as 0.302 out of 1.0, rather than plain magnitude’s near-zero 0.018, and the early-fusion Dense layer sees this elevated background everywhere. This version produced the largest regression of the entire investigation, including on seated recall, the thing it was meant to fix. Since normalizing is a binary choice and both branches were tried and failed for these opposite reasons, this line of investigation is closed.

A per-cell d' is a feasibility check, not a prediction. The coherence channel delivered 177.9 int8 levels of seated contrast against plain magnitude’s 7.4 — a 24× better per-cell feature — and lost by 8–11 paired SE. Every gate it was given passed, several spectacularly. The number that predicted its failure was printed by the same gate table and not thresholded on. A printed number nobody asserts on is documentation, not a gate.

7.3 What did work: the decode, not the input

Every approach above tried to make a static body visible within a single frame. What actually worked instead operates on the model’s own output, using a property none of the input-side attempts could exploit: a seated person occupies the same grid cell for the entire recording, while a spurious detection does not.

The heatmap is smoothed with a causal EMA ($\tau = 10$ s) and decoded as $\max(P, S \cdot \theta/t_{10})$ at the unchanged threshold θ , so a cell fires if it is bright now **or** has averaged above t_{10} for a while. The rescaling lets one threshold decode both cases on a single surface, which keeps the NMS and centre-of-mass steps seeing a coherent peak. It costs one float32 state array of 864 bytes and no retraining.

The diagnosis behind this splits the “seated problem” into two distinct causes. For each seated person we can ask whether the per-frame decode recalled them, and separately, how their own grid cell ranked among all 216 cells of that frame’s heatmap — a rank near the top means the model’s raw output already pointed at the right place, even though the decode’s threshold discarded it:

Condition	Seated recall	Rank of the seated cell of 216
walker 1.5–2.0 m away	64.6%	median 15
walker 2.0–3.0 m away	64.2%	(chance would be 108)
walker 3.0–9.0 m away	60.0%	—
no walker in the room (w021)	0.6%	median 203 — bottom 6%

In **mixed** windows the model does place real evidence at the seated cell and the per-frame decode was discarding it; that is 76% of all seated misses, and it is what persistence recovers. In w021 the cell ranks below 94% of the room — there is nothing to decode, and no output-side method can reach it.

This is a genuine trade, not a free improvement. Persistence gains +0.0184 F_1 on the eight seated-containing windows and costs 0.0026 on the fourteen walking-only ones. The net pooled gain therefore depends on the proportion of seated-subject recordings resembling that of the training data (roughly one third of windows here); a session with substantially fewer such occurrences would see a correspondingly

smaller benefit. An obvious refinement — gating the promotion on how steady a cell’s activation is, to distinguish a stationary subject from a moving person’s decaying trail — was tried, and restores walking-only F_1 exactly as fast as it removes the seated gain.

8 Results

Cross-validation F_1 : 0.8682 ± 0.018 (1 SE), threshold 0.50, scored on the deployed int8 path with conditional inference and persistence both active. This is the first out-of-fold measurement taken on int8 output rather than float Keras output — the two paths peak at different thresholds (int8 0.50, float 0.40) and disagree by $+0.0145 F_1$ at their respective optima. Measured seated recall is 68.6%, consistent with the 68.1% figure obtained independently on a separate machine.

Constraints: 349.5 KB / 800 KB flash (44%), 11 operators, no forbidden ops, full INT8 — 5/5 checks pass. The activation-arena estimate reads **26–57 KB of the 300 KB budget** depending on the machine: it is a heuristic over the interpreter’s tensor plan and moves with the TensorFlow build. Three runs on each of our two machines gave 56.6–57.4 KB and 26.2–27.0 KB (TF 2.20.0) for the same model file. Either reading leaves a 5–11 $\times$ margin, and the flash figure reproduces identically on both.

Conditional inference: 95.1% of empty-room frames skipped, 0.0000% of occupied frames skipped across every window where a subject is separable from empty at all, 11.5% of all frames pooled over the 24-window set.

Post-processing: a full grid search (1008 configurations) over the decode threshold, NMS suppression radius, and tracker smoothing/coasting was run against this cache. It found a configuration reaching $F_1 = 0.8820$, $+0.0138$ over the deployed one — but that result sits on the boundary of the searched threshold range and was not verified with a paired SE against the deployed configuration. We therefore kept the original constants ($\theta = 0.50$, tracker $\alpha = 0.4$, $\text{max_distance} = 1.0$, $\text{max_coast} = 2$) rather than adopt an unverified result.

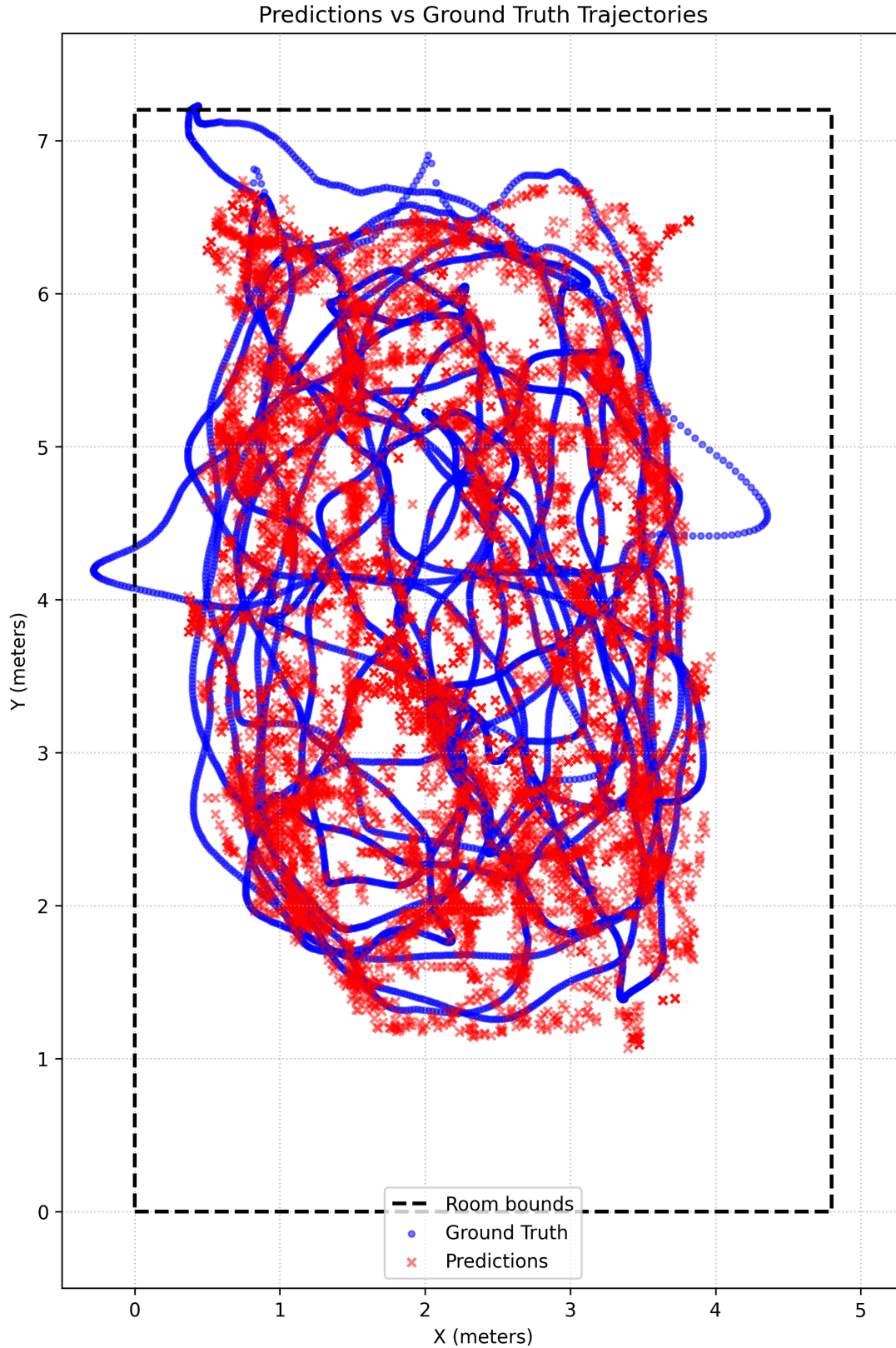


Figure 3: Decoded occupancy against ground-truth positions. Peaks are refined to sub-cell precision by the centre of mass of their 3×3 neighbourhood before non-maximum suppression.

9 A documented negative: runtime layer skipping

Dynamic depth — skipping layers per frame according to input difficulty — is the natural extension of Section 6 from the frame level into the graph, and it cannot ship here. TFLM plans its activation arena **statically** at

`allocate_tensors()` and forbids CUSTOM operators, so there is no supported way to vary the executed subgraph at runtime. Conditional inference therefore delivers its saving entirely outside the graph, in the host code that decides whether to invoke the interpreter at all — which violates nothing and needs no operator support.

10 Conclusion

The system meets every hard constraint with a wide margin and reaches its accuracy through a small number of changes that were each measured against a leave-window-out estimate and an explicit adoption bar. The two findings most worth carrying forward are methodological rather than architectural: that an optimization must be validated on the metric it can break, not the one it was aimed at (Section 4), and that a summary statistic measured away from the decision boundary — a per-cell d' , a pooled separability — repeatedly failed to predict behaviour at it (Sections 6 and 7).